

Assignment 5. Bisection & Regula Falsi

Marks 20

Posted on 02.09.2026 @ 2:30 pm and due on 02.09.2026 @ 5:40 pm

1. Code up library functions Bisection and Regula Falsi. **[5+5]**
2. Find the root of the following function in the interval $[1.5, 3.0]$ to an accuracy of 10^{-6} using both Bisection and Regula Falsi method. **[3+3]**

$$f(x) = \log(x/2) - \sin(5x/2)$$

3. Consider the following non-linear equation

$$f(x) = -x - \cos(x)$$

Find an appropriate interval bracket in which the root possible is, starting with the interval $[2, 4]$. **[4]**